

Topological Causation in Lattice QCD: A Reading of the Field’s Own Results

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“It is fun to speculate about the way quarks would behave if they were physical particles of finite mass (instead of purely mathematical entities as they would be in the limit of infinite mass).” — M. Gell-Mann, Phys. Lett. 8, 214 (1964)

Abstract

We note a recurring pattern in the lattice QCD literature concerning the mechanism of color confinement and the status of topological charge. The field’s operational practice—causal experiments of vortex removal and vortex-limited Wilson loops, continuum extrapolations, action-independence checks, and algorithmic responses that navigate rather than dissolve topology—treats center vortices and topological charge as physical features of the theory. Its interpretive language, however, continues in places to frame the same structures with a residual provisionality. The field’s canonical review of the confinement program, published in 2003, already located the real argument for the physical content of center projection in the empirical correlations between lattice-identified objects and gauge-invariant observables. In the two decades since, the empirical results have continued to accumulate and strengthen. The same structural mismatch between operational practice and interpretive framing appears, in a different register, in the contemporary literature on topological freezing. The passages examined below are selected as clear illustrations of the language under discussion; they are not offered as a statistically representative sample of the full literature. The pattern is documented on the field’s own terms, in the field’s own notation, and in three distinct vocabularies. The contribution of the present work is the explicit identification of this practice/rhetoric mismatch. The interpretive question is left to the field.

1 Introduction

The lattice formulation of quantum chromodynamics provides a non-perturbative regulator that renders the theory calculable. Space-time is discretized into a grid with spacing a , and physical results are recovered in the continuum limit $a \rightarrow 0$. Within this framework, numerical simulations have identified specific topological structures—center vortices and topological charge—that correlate with the non-perturbative phenomena of quark confinement and chiral symmetry breaking.

The question of whether these structures are physical features of the theory or artifacts of the discretization has been addressed repeatedly in the literature. The present work does not propose a new framework, introduce new mathematics, or take any particular stance on their

physical status. It notes a pattern: the field’s operational practice has long treated the topological structures as physical—via the causal experiments, continuum limits, and action-independence tests that constitute its standard methodology—while its interpretive language has, in places, continued to qualify those same structures as though their status were still provisional. A conclusion stated by the field’s canonical review in 2003 has been continuously supported by subsequent empirical work.

What follows is strictly a reading of the field’s own materials. Section 2 establishes the relevant mathematics in standard notation. Section 3 presents the argument as the canonical review itself stated it in 2003, together with the pattern of interpretive language that has persisted since, while distinguishing obligatory technical caveats from any residual interpretive gap. Section 4 examines the contemporary literature on topological freezing as a second, contemporary illustration of the same structural mismatch. Section 5 names the cumulative pattern—the mismatch between operational practice and interpretive rhetoric—addresses the universality objection, and draws the methodological consequence that follows from the field’s own vocabulary of “tricks.” Section 6 states precisely what has been shown and what has not.

The paper is written so that it can stand alone while also serving as the third vertex of a triangle formed with a three-way constraint taxonomy of the fundamental interactions and a category-error analysis of the black-hole information paradox.

2 Mathematical Preliminaries

2.1 The Wilson loop and the area law

In lattice gauge theory the fundamental gauge-invariant observable is the Wilson loop

$$W(C) = \frac{1}{N} \text{Tr} \mathcal{P} \prod_{x \in C} U_\mu(x), \quad (1)$$

where C is a closed curve on the lattice, $\mathcal{P}$ denotes path ordering along C , and the trace is taken in the fundamental representation of $SU(N)$. For large rectangular loops with area A , the expectation value in the confined phase obeys an area law:

$$\langle W(C) \rangle \sim e^{-\sigma A}, \quad (2)$$

where σ is the asymptotic string tension. This linear potential is the lattice signature of quark confinement: the energy required to separate a quark–antiquark pair grows with distance, preventing isolation of color charges.

The string tension $\sqrt{\sigma}$ is a physical quantity that can be determined from lattice simulations and extrapolated to the continuum. Its persistence across different lattice actions and into the continuum limit is a central empirical fact of the confinement program.

2.2 Center projection and P-vortices

To isolate the degrees of freedom responsible for the area law, the link variables $U_\mu(x)$ are fixed to a maximal center gauge and projected onto the center $\mathbb{Z}_N$ of the gauge group:

$$U_\mu(x) = Z_\mu(x) V_\mu(x), \quad Z_\mu(x) \in \mathbb{Z}_N. \quad (3)$$

The projected configuration $Z_\mu(x)$ identifies the locations of thin center vortices (P-vortices) on the dual lattice. The center-projected Wilson loop $Z(C)$, evaluated on the projected links alone, reproduces the asymptotic string tension of the full theory to within a few percent. This phenomenon is known as center dominance and has been reproduced across multiple gauge-fixing procedures and lattice actions since the late 1990s.

2.3 Vortex-removed and vortex-limited configurations

The causal role of the identified vortices is tested by two complementary constructions. The vortex-removed configuration is obtained by

$$V_\mu(x) = Z_\mu^*(x) U_\mu(x). \quad (4)$$

Wilson loops evaluated on $V_\mu(x)$ exhibit a perimeter law rather than an area law: removal of the P-vortex content destroys the confining behavior of the original ensemble. Conversely, vortex-limited loops $W_n(C)$ are evaluated on configurations through which a definite number n of P-vortices passes. These restricted loops recover the area-law dependence on n predicted by a vortex picture of confinement.

These experiments—vortex removal destroying confinement and vortex limitation restoring it—constitute the empirical core of the lattice argument that center vortices are causally responsible for the area law.

2.4 Topological charge and topological freezing

The topological charge of a gauge configuration is

$$Q = \frac{g^2}{32\pi^2} \int d^4x F_{\mu\nu}^a \tilde{F}^{a\mu\nu}, \quad (5)$$

where $\tilde{F}$ is the dual field strength and a is the adjoint color index. On the lattice the definition of Q is prescription-dependent at finite spacing: it depends on the choice of smoothing or gradient-flow time, small instantons may fall through the lattice, and strict integer-valuedness requires an admissibility condition. These are genuine practical difficulties of the finite- a theory.

In the continuum limit the ambiguities are removed. Gradient-flow definitions of the topological charge become unambiguous as $a \rightarrow 0$, and continuum-extrapolated values of the topological susceptibility $\chi_t = \langle Q^2 \rangle / V$ are standard (see, e.g., the FLAG reviews [12]). Once continuum-extrapolated, χ_t is a physical, gauge-invariant observable that enters the Witten–Veneziano relation and contributes to the η' mass. No serious claim has been advanced that the continuum topological charge is an algorithmic illusion.

Standard local-update Monte Carlo algorithms sample gauge configurations by a Markov process. The efficiency of the process for the topological charge is characterized by an integrated autocorrelation time τ_Q , which grows sharply as the lattice spacing is reduced. As $a \rightarrow 0$, the relation

$$\tau_Q \sim a^{-z} \quad (6)$$

holds with a positive exponent z that is empirically large; in standard hybrid Monte Carlo $z \approx 5$. The Markov chain becomes progressively less able to move between topological sectors, and in the approach to the continuum limit it remains trapped in a single sector for the duration of the simulation. This phenomenon is known as topological freezing. Algorithmic responses including

open boundary conditions and parallel tempering on boundary conditions (PTBC) mitigate the freezing. These responses improve the sampling of topological sectors without dissolving the topological structure itself; the topology remains a feature that the improved algorithms must navigate.

3 The Argument as the Canonical Review Stated It

The lattice case for the causal role of center vortices in confinement rests on the body of vortex-removal and vortex-limited experiments accumulated over more than two decades. We summarize the load-bearing results in the field’s own notation and note a recurring feature of the interpretive language used to describe them. The passages examined are selected as clear illustrations of the language under discussion; they are not offered as a statistically representative sample of the full literature.

3.1 The empirical claim

The decomposition (3) and the vortex-removed construction (4) of Section 2 define the experimental apparatus. The empirical results obtained with it are threefold. First, center dominance: the center-projected Wilson loop $Z(C)$, evaluated on the projected configuration alone, reproduces the asymptotic string tension of the full theory to within a few percent, a result reported in the late 1990s and reproduced since across multiple gauge-fixing procedures and lattice actions. Second, vortex removal: Wilson loops evaluated on the vortex-removed configuration $V_\mu(x)$ exhibit a perimeter law rather than an area law, so that removal of the P-vortex content destroys the confining behavior of the original ensemble. Third, vortex limitation: loops $W_n(C)$, restricted to configurations through which a definite number n of P-vortices passes, exhibit the area-law dependence on n predicted by a vortex picture of confinement. These are the experiments which have for two decades anchored the lattice argument that center vortices are causally responsible for the area law.

3.2 The interpretive language

The interpretive language with which the field describes the objects identified by center projection is consistent across decades and across independent authors in one respect, and heterogeneous in another. Representative passages illustrate the range.

The original reports of center dominance characterize the result in directly causal terms [1, 2]:

We find that center vortices identified in the vacuum state account for the full asymptotic string tension.

The review which superseded them characterizes the same result in more guarded terms [3]:

We review lattice evidence showing that center-vortex condensation is a serious candidate for the mechanism of colour confinement in quantum chromodynamics.

The canonical review of the field’s confinement program introduces the same body of results with stronger language [4]:

[T]he prevailing view is that quark confinement is the work of some special class of gauge field configurations—candidates have included instantons, merons, abelian

monopoles, and center vortices—which for some reason dominate the QCD vacuum on large distance scales. . . the underlying mechanism of quark confinement, via definite classes of field configurations, is open to numerical investigation.

The same review, in a methodologically careful aside, draws a distinction between the lattice-identified objects and the physical objects they are taken to represent [4]:

It is obvious that the action of thin vortices, stable or not, is singular in the continuum limit, and their contribution to the functional integral can be neglected. The configurations of physical interest are vortices which have some finite thickness d in physical units.

A contemporary paper from the most active vortex-causation group restates a related two-level distinction [10] (internal citations omitted):

Although the identified thin vortices are gauge dependent, they do correlate with the physical thick vortices and it is possible to learn about the latter by studying the former.

Part of this two-level language is technically obligatory. Center projection is a gauge-fixing procedure; the resulting P-vortices are gauge-dependent by construction. Any careful paper must register the distinction between the gauge-fixed identification and the gauge-invariant physics it is taken to locate. The 2024 formulation just quoted is a clear instance of this obligatory bookkeeping; the same paper’s title (“Numerical indication that center vortices drive dynamical mass generation in QCD”) already states the causal claim without hesitation. Not every restatement of the thin/thick distinction is therefore evidence that the relationship between the two levels remains open.

What remains visible, however, is a recurring tendency in some of the literature to present the empirical correlations themselves in language that continues to treat the physical status of the underlying configurations as more provisional than the 2003 *Critique* had already allowed. The real argument, according to that *Critique*, resides in the causal experiments and the correlations with gauge-invariant observables.

3.3 The argument as the canonical review states it

The most direct statement of the position the field has arrived at appears not in any of the affirmative passages above, but in the section of the canonical review explicitly devoted to objections. In Section 6.7, titled simply *Critique*, the author of the review reviews the standard methodological complaints against the center-projection program—the gauge-fixing dependence of the projection, the variation between adjoint gauges, the sensitivity to Gribov copies—and writes the following [4]:

[T]he real argument that center projection locates vortex configurations is at present empirical; namely, the correlation of P-vortices with gauge-invariant observables, as seen in the results for vortex-limited Wilson loops $W_n(C)$, and the observables of vortex-removed configurations $V_\mu = Z_\mu^* U_\mu$.

Three features of this passage are salient. First, the author of the field’s canonical review identifies the real argument for the physical content of center projection as residing in the causal experiments and in the correlation of the lattice-identified objects with gauge-invariant

observables. Second, this statement appears in the section devoted to addressing methodological objections; it is not an enthusiastic claim made in passing but a considered statement of where the case rests after the objections have been heard. Third, the passage was published in 2003. It has not been subsequently retracted, qualified, or revised in the literature that draws on the review.

The position the field’s canonical review reached two decades ago is therefore the following: the gauge-fixing apparatus of center projection is a calculational scheme whose own structural features are not the object of physical claims; the physical claim is that this scheme correlates with gauge-invariant observables and with causal experiments in a manner that licenses inference about the underlying configurations. This is a structurally careful position. We do not contest it.

3.4 A converging statement from outside the projection program

The position arrived at in the canonical review is not isolated. J. M. Cornwall, writing in 2009 from outside the center-projection methodological tradition, reaches the same conclusion in directly substantive terms [6]:

Center vortices have been around for more than thirty years, well-confirmed on the lattice, and very successful in explaining the basics of confinement. . . basic confinement in the center vortex picture is topological, coming from gluonic solitons where the gluons have no direct coupling to the Wilson loop, and makes no reference to any particular surface.

This passage commits to the topological character of the confinement mechanism without the intermediate gauge-fixing apparatus that has been the subject of methodological caution in the projection literature. The substantive content of the position the canonical review reached through its empirical argument is the substantive content of the position Cornwall articulates directly. The two routes converge on a single claim about the underlying mechanism.

3.5 Noted tension

The pattern we note in the present section is therefore narrower than a blanket claim of non-uptake. The lattice case for the causal role of center vortices rests on two decades of vortex-removal and vortex-limited experiments. In 2003 the field’s canonical review, in a section devoted to methodological objections, located the real argument for the physical content of center projection precisely in those experiments and in the correlations with gauge-invariant observables.

Some of the two-level language that has persisted since is the ordinary and necessary distinction between gauge-fixed identifications and gauge-invariant physics; that distinction does not dissolve and is not, by itself, a failure to integrate the 2003 conclusion. Other formulations, however, continue to present the relationship between the lattice-identified objects and the physical configurations as more open than the empirical case—and the 2003 statement of where that case rests—already supports.

We note this more limited tension. We do not, in this section, draw further conclusions from it. Section 4 develops the analogous pattern in the contemporary literature on topological freezing. The cumulative observation across both domains is the object of Section 5.

4 Illustration: Topological Freezing and Its Algorithmic Responses

The vortex-causation evidence summarized in Section 3 is now more than two decades old. The literature on topological freezing, which has intensified since approximately 2017 and remains active in 2024, provides a contemporary illustration of a related structural pattern. It is important to be precise about the epistemic status of the quantities involved. Unlike the identification of center vortices, the physical reality of the continuum topological charge and of χ_t has never been seriously contested: χ_t is gauge-invariant, enters the Witten–Veneziano relation, and contributes to the η' mass. The present section therefore does not adjudicate any question of physical status; it illustrates, in a different and more recent register, the same mismatch between operational practice and interpretive framing already visible in the vortex literature.

4.1 The phenomenon

As the continuum limit is approached ($a \rightarrow 0$), the autocorrelation time τ_Q associated with the topological charge diverges. Standard Monte Carlo algorithms become trapped in a single topological sector and are unable to tunnel between sectors with different values of Q . This topological freezing constitutes a strict algorithmic obstruction. Responses such as parallel tempering on boundary conditions (PTBC) have been developed to mitigate the freezing. These responses improve sampling without dissolving the topological structure itself; the topology remains a feature that the improved algorithms must still navigate.

4.2 The interpretive language

A recent PTBC study of scale setting in $SU(N)$ Yang–Mills theory illustrates the pattern. In its conclusions the authors write [11]:

Our preliminary analysis shows that, in the presence of a completely frozen topology, t_0 receives a positive bias with respect to the actual value obtained with a properly sampled topological charge.

Three phrases in this single sentence do substantive work. “Completely frozen topology” treats the topology as a feature that can be frozen or unfrozen. “The actual value” presupposes that there exists a fact of the matter independent of any particular sampling. “Properly sampled topological charge” presupposes that sampling can be done correctly or incorrectly with respect to that fact. The sentence as a whole therefore commits, implicitly but unambiguously, to the topological charge being a real physical quantity that possesses values and that can be sampled properly or improperly. The same paper frames the PTBC method throughout as an algorithmic improvement that addresses a “well-known problem of standard algorithms in the sampling of the topological modes.” Both registers—substantive commitment to a physical topological feature and algorithmic framing of the difficulty—operate simultaneously.

4.3 The phrase “properly sampled”

The commitment implicit in “properly sampled” is significant. One cannot sample an algorithmic illusion properly or improperly; one can only sample a quantity that possesses an independent ground truth. By identifying their result as a bias relative to the actual value obtained with proper

sampling, the authors indicate that they are attempting to measure a real, independent quantity. A practitioner may reply that proper sampling refers only to ergodicity of the Markov chain with respect to the lattice-defined ensemble at fixed a , a reading which requires no commitment beyond the mathematics of the algorithm. The reply is correct as far as it goes; the weight of the observation accordingly rests not on this phrase alone but on the broader practice/rhetoric mismatch developed in Section 5, to which the phrase is a contemporary witness.

4.4 Illustration noted

The contemporary topological-freezing literature therefore exhibits, in a different register, the same structural pattern documented in Section 3 for the vortex-causation literature. In both cases the substantive content of the work treats a topological feature of the theory as something that can be sampled correctly or incorrectly, or whose causal role can be tested by removal and restoration experiments, while the interpretive language continues in places to frame the feature in primarily methodological or provisional terms. Two areas of lattice research, separated by two decades, exhibit a related pattern. Section 5 names that pattern and draws its methodological consequence.

5 The Pattern and Its Implications

We have noted, in Section 3, that the case for the causal role of center vortices in confinement was stated by the field’s canonical review in 2003, in a section explicitly devoted to addressing methodological objections, in terms which locate the real argument in the empirical correlations between the lattice-identified objects and gauge-invariant observables. We have noted, in Section 4, that the contemporary literature on topological freezing exhibits a related structural pattern in a different register. We turn now to that pattern.

5.1 The methodological vocabulary and the universality response

A familiar word is used, across the lattice literature, to characterize the methodological status of the lattice itself: it is a *trick*. The word is also used, in the technical vocabulary of theoretical physics, of Wick rotation, of dimensional regularization, of the Faddeev–Popov procedure, of the $i\epsilon$ prescription. In those uses the word is precise and unobjectionable. Each refers to a calculational device whose structural features are introduced to make a problem tractable and whose specific structural features do not survive into the final answer. One rotates time into the imaginary axis, computes, rotates back; the imaginary time is not present in the result. One treats the spacetime dimension as a continuous variable, isolates the divergence, takes the limit; the non-integer dimension is not present in the result. The trick, in each case, vanishes when the calculation is complete.

The lattice’s topological content does not vanish. To wit: the topological charge Q , the topological susceptibility χ_t , the causal role of center vortices in the area law, the persistence of $\sqrt{\sigma}$ across independent actions and into the continuum limit, the algorithmic obstruction posed by topological freezing as $a \rightarrow 0$ —none of these is a structural feature of any particular

discretization that disappears when the discretization is removed.¹ They are features that the discretization makes accessible to computation, and they remain in the answer. By the criterion implicit in the field’s standard use of “trick” for Wick rotation and its kin, that which survives the removal of the calculational scaffolding is not part of the scaffolding.

A lattice practitioner will immediately recognize this criterion as the operational definition of universality. The Wilsonian program is organized precisely around the survival test: one varies the discretization (action, spacing, boundary conditions), takes the continuum limit, and retains only those features that persist independently of the particular scaffolding. Action-independence of the string tension, continuum extrapolation of χ_t , and the persistence of center dominance across gauges are the field’s own implementations of the test. The referee’s one-line reply writes itself: the author has rediscovered why we take the continuum limit.

We concede the point without reservation. The survival criterion articulated above is not a novel methodological insight; it is the field’s standard practice. The contribution of the present work therefore cannot lie in the criterion itself. It lies in the observation that the field’s interpretive register has not, in every respect, kept pace with its operational practice.

Operationally, the field treats the topological structures as physical. It designs causal experiments (vortex removal, vortex-limited loops) whose outcomes are interpreted as evidence of causal responsibility for the area law. It invests in algorithmic responses (PTBC, open boundaries) that navigate rather than dissolve the topology. It reports continuum limits and action-independent results for quantities whose physical content is thereby established by the very universality tests the field applies. Rhetorically, the same literature continues in places to frame the lattice-identified objects with a residual provisionality, or to characterize the lattice itself as a “trick” even while its topological content survives the removal of the scaffolding. The practice/rhetoric mismatch is the pattern documented in Sections 3 and 4; the methodological vocabulary of “trick” merely restates it in a third register.

The vocabulary of instrumental deferral is older than the lattice itself. The 1972 proceedings contribution in which the color-octet gluon scheme later named QCD was first proposed opens by describing a field-theory model of the hadrons that its authors “don’t fully believe,” titles its second section “Fictitious Quarks and ‘Gluons’ and Their Statistics,” and closes by proposing a complete theory of hadrons built on constituents “treated as fictitious” [14]. The same document cites the SLAC deep-inelastic results as its motivation, and grounds the fictitious framing in the assumption that quarks are “permanently bound inside the mesons and baryons,” and refers to the nine colored quarks as “mathematical entities.” Unlike the imaginary time of Wick rotation, the entities did not vanish from the answer; the mechanism of their permanent binding is the topological structure whose causal status the preceding sections document. We note this history as register, not as evidence; it predates the corpus examined here.

5.2 Three statements of one observation

We note that this consequence is not separable from the pattern documented in Sections 3 and 4. It is the same observation—the mismatch between what the field computes and how it

¹Analytic continuum realizations of the corresponding defects, motivated by the lattice identification methods, appear in Asmaee, Deldar, and Kiamari [15], who construct center vortices (and, via an Abelian gauge step, monopole–vortex chains) directly in the continuum QCD vacuum using the connection formalism, without discretization. We note the construction as a continuum witness to the survival criterion, not as independent evidence for the causal claims of Sections 3 and 4; it employs the same gauge-fixing apparatus whose obligatory bookkeeping is discussed in Section 3.2, and relocates rather than dissolves the thin/thick distinction.

sometimes talks—restated in the language of the methodological vocabulary the field has used to characterize its own procedures. The canonical review’s location of the real argument in the empirical correlations is the same observation again: the empirical correlations are precisely the features that survive the calculational procedure, and the review’s 2003 statement already treated them as settling the relevant physical question. The contemporary topological-freezing literature’s commitment to a properly sampled topological charge is the same observation again: a feature that admits proper or improper sampling is a feature with content of its own. Three statements of one observation, separated by two decades and two areas of research, appear in three distinct vocabularies.

A compressed pedagogical exhibit of the same observation appears in a 2021 introductory lecture by Michael Creutz. Presenting Wilson’s 1973 strong-coupling lattice theory, Creutz writes on a single slide [13]:

Strong coupling limit does confine quarks

- only quark bound states (hadrons) can move

space-time lattice = non-perturbative cutoff
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Lattice gauge theory

- A mathematical trick
- Minimum wavelength = lattice spacing a
- Maximum momentum = π/a

Be discrete, do it on the lattice

Be indiscreet, do it continuously

The slide places in immediate proximity the strong-coupling result that the lattice confines quarks, the identification of the lattice itself as a non-perturbative cutoff, the characterization of lattice gauge theory as a mathematical trick, and the pedagogical injunction to be discrete on the lattice while remaining continuous in the physical theory. The slide therefore compresses, onto a single piece of pedagogical material in the field’s most authoritative voice, the same observation: the lattice is a calculational device in which the topological content survives into the answer.

The pattern across these three statements—Greensite (2003), Bonanno et al. (LATTICE2024), and Creutz (2021)—is consistent in its substantive content and uneven in its interpretive articulation. The substantive content is that the topological structure of QCD, as manifested on the lattice and surviving into the continuum, is a feature of the theory whose physical relevance is established by the empirical correlations, causal experiments, and continuum extrapolations that the field has been accumulating. The interpretive articulation has, in places, continued to qualify this content in provisional terms. The recurrence of that residual gap, in the face of operational practice that has long since settled the relevant questions, is the pattern we have been noting.

5.3 An observation in closing

We close with one further observation, addressed to the lattice community in particular. The conclusion the canonical review reached in 2003 is the conclusion the empirical work since 2003 has continued to support and strengthen. The vortex-removal experiments are more refined; the

action-independence is more thoroughly established; the algorithmic responses to topological freezing now demonstrate that the topology must be navigated rather than dissolved. The argument has not weakened. The operational practice has been consistent with that conclusion for two decades. What remains, in places, is for the interpretive register to catch up. We note, in the spirit of recovery rather than imposition, that the conclusion the canonical review’s empirical case supported is available to the field on its own terms.

6 Conclusions

What we believe has been shown is the following. The lattice literature contains a body of empirical results—vortex-causation experiments accumulated since the late 1990s, contemporary work on topological freezing and its algorithmic responses—whose interpretive consequences were stated by the field’s canonical review in 2003. The subsequent literature has not, in every respect, integrated those consequences into its interpretive framing in a manner fully consistent with the strength of its own operational practice. This pattern is documented in Sections 3 and 4 and named, together with its methodological consequence and the universality concession, in Section 5. The pattern appears across two decades and two areas of lattice research. Its lasting formulation is the meta-observation that the field’s interpretive register has not, in every respect, kept pace with its operational practice.

What has not been shown is the following. No specific interpretive commitment beyond the operational evidence has been asserted as correct. No claim has been made that the standard interpretation is wrong in a manner that affects calculational outputs. No extension has been proposed beyond the domain of QCD. The survival criterion of Section 5.1 is conceded to be the field’s own universality practice; the contribution is the identification of the residual mismatch between that practice and some of the language in which the results continue to be framed.

The interpretive question is left to the field.

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